

Lab 10 - Blockchain Invoice Verification

Simulate blockchain-style invoice immutability using a Flask REST API running in Docker. Register an invoice, verify it against a SHA-256 hash, and detect tampering.

Lab environment: Docker Desktop (local machine) - required for this lab Estimated time: 30 minutes

Overview

Endpoint	Method	Purpose
/health	GET	Check service status
/register	POST	Register invoice and store its hash
/verify	POST	Verify invoice - detect tampering
/invoices	GET	List all registered invoices

Step 1 - Clone the Lab 10 source from GitHub

```
git clone https://github.com/tertiarycourses/TGS-2026064859-Autonomous-AI-Agents-with-OpenClaw.git
cd TGS-2026064859-Autonomous-AI-Agents-with-OpenClaw/labs/lab-10-blockchain-invoice
```

Step 2 - Build the Docker image

```
#docker build -t openclaw/invoice-verify:latest .
docker build -t invoice-verify:1.0 .
```

Expected output: `` Successfully built xxxxxxxxxx Successfully tagged invoice-verify:latest ``

Step 3 - Run the blockchain verification container

```
#docker run -d -p 5000:5000 --name invoice-verify \
  openclaw/invoice-verify:latest

docker run -d \
  --name invoice-verify \
  -p 5000:5000 \
  invoice-verify:1.0 \
  sleep 2 \
  curl http://localhost:5000/health
```

Expected output: ``json {"invoices": 0, "status": "ok"} ``

Step 4 - Register a test invoice

```
curl -X POST http://localhost:5000/store -H "Content-Type: application/json" -d '{"invoice_id\":"INV-2026-001\","vendor\":"Tertiary Infotech\","amount\":"1500.00\","date\":"2026-07-11\":"}'
```

Expected output: `` {"status":"stored"} ``

Step 5 - Verify the invoice - same data (should pass)

```
curl -X POST http://localhost:5000/verify \
-H 'Content-Type: application/json' \
-d '{"invoice_id":"INV-2026-001","vendor":"Tertiary Infotech",
  "amount":1500.00,"date":"2026-07-11"}'
```

Expected output: ``json {"invoice\_id": "INV-2026-001", "verified": true, "tampered": false} ``

Step 6 - Tamper test - change amount (should fail)

```
curl -X POST http://localhost:5000/verify \
-H 'Content-Type: application/json' \
-d '{"invoice_id":"INV-2026-001","vendor":"Tertiary Infotech",
  "amount":9999.00,"date":"2026-07-11"}'
```

Expected output: ``json {"invoice\_id": "INV-2026-001", "verified": false, "tampered": true} ``

The SHA-256 hash of the tampered data does not match the stored hash - simulating blockchain immutability.

step 7 -Now perform the tamper test

Change the amount from 1500.00 to 9999.00:

```
curl -X POST http://localhost:5000/verify \
-H "Content-Type: application/json" \
-d '{"invoice_id\":"INV-2026-001\","vendor\":"Tertiary Infotech\","amount\":"9999.00\","date\":"2026-07-11\":"}'
```

Expected output:

```
{"invoice_id":"INV-2026-001","tampered":true,"verified":false}
```

Verification

Check	Expected
curl http://localhost:5000/health	{"invoices": 1, "status": "ok"}
Verify with correct data	"verified": true, "tampered": false
Verify with wrong amount	"verified": false, "tampered": true

Cleanup

```
docker stop invoice-verify
docker rm invoice-verify
```

Reference

- Lab source code: <https://github.com/tertiarycourses/TGS-2026064859-Autonomous-AI-Agents-with-OpenClaw/tree/main/labs/lab-10-blockchain-invoice>
- Flask docs: <https://flask.palletsprojects.com>